## Abstract

Two approaches were followed in order to detect a broader number of chemicals in the context of the CASMI 2022 exercise. On the one hand, we used MS-DIAL and all the public spectral libraries available at the time we performed the analysis (see methods section for details). Moreover, the peaks were imported to MS-FINDER, i.e., when no match in MS-DIAL libraries was found, in silico fragmentation was performed with MS-FINDER. On the other hand, patRoon software was employed to perform the same analysis, as we noticed in previous studies, that MS-DIAL and patRoon results could be complementary.

For simplicity, we have elaborated two different tables; one containing the results derived from MS-DIAL + MS-FINDER software, and the other one containing the results we obtained from patRoon.

## Methods

### MS-DIAL

The mzML files provided were uploaded to MS-DIAL (version 4.80). For the annotation, the latest MSP libraries available were used: in particular, "MSMS-Public-Pos-VS16.msp" and "MSMS-Public-Neg-VS15.msp" were implemented for the positive and negative ionization modes, respectively. No peak alignment was performed. The parameters used for the data analysis are summarized in Table 1.

*Table 1. Parameters selected for the MS-DIAL data analysis.*

| Parameter settings MS-DIAL | | |
|---|---|---|
| **Data collection** | | |
| **Parameter** | **Value for analyses in positive** | **Value for analyses in negative** |
| **MS 1 tolerance** | 0.01 Da | 0.01 Da |
| **MS 2 tolerance** | 0.025 Da | 0.025 Da |
| **Retention time begin** | 0 min | 0.5 min |
| **Retention time end** | 100 min | 10 min |
| **MS1 mass range begin** | 0 Da | 0 Da |
| **MS1 mass range end** | 2000 Da | 2000 Da |
| **MS/MS mass range begin** | 0 Da | 0 Da |
| **MS/MS mass range end** | 2000 Da | 2000 Da |
| **Maximum charged number** | 2 | 2 |
| **Peak detection** | | |
| **Minimum peak height** | 1000 amplitude | 100000 amplitude |
| **Mass slice width** | 0.05 Da | 0.05 Da |
| **Smoothing level** | 3 scan | 3 scan |
| **Minimum peak width** | 5 scan | 5 scan |
| **MS2 deconvolution** | | |
| **Sigma window value** | 0.5 | 0.5 |
| **MS/MS abundance cut off** | 0 amplitude | 0 amplitude |
| **Identification** | | |
| **Retention time tolerance** | 100 min | 0.5 min |
| **Acurate mass tolerance (MS1)** | 0.01 Da | 0.005 Da |
| **Acurate mass tolerance (MS2)** | 0.05 Da | 0.01 Da |
| **Identification score cut off** | 80% | 80% |
| **Adduct** | | |
| **Adducts** | [M+H]+, [M+NH4]+, [M+Na]+, [M+H-H2O]+ | [M-H], [M-Cl]-, [M-H-H2O]- |

**patRoon**

The provided mzML files were directly uploaded to the software and features were extracted and grouped with *XCMS3*. Then the relevant MS data was extracted using the *mzR* algorithms. Molecular formulas were calculated using *GenForm* and compound identification was performed with MetFrag (MetFrag2.4.5-CL jar file) and the PubChem database. When multiples options were found for a particular m/z of interest, the feature with the higher fragScore was selected; if multiple features were tied for the fragScore, the generic score was considered.

Even though we report only one annotated compound per feature using the above-mentioned score prioritisation system, we also have the full dataset of compounds and we are willing to compare if the correct result is still present among them, to try to understand and eventually to improve the score system.